9.) $G = \mathbb{R} \setminus \{-1\}$. Chap 9 Isomorphisms

In chap 3 exercises we proved that G is a group w.r.t the operation

$$a * b = a + b + ab$$

The identity element in G is $e = 0$

as $\forall a \in G \quad a * 0 = a + 0 + a \cdot 0$
$= a$

The inverse a^{-1} is given by

$$a^{-1} = \frac{-a}{1+a}$$

Claim $(G, *) \cong \mathbb{R}^* = \{x \in \mathbb{R} : x \neq 0\}$ under multiplication.

We must find an isomorphism

$$\phi : G \to \mathbb{R}^*$$

ie a bijective map ~~that~~ that satisfies.

$$\forall a, b \in G \quad \phi(a * b) = \phi(a)\phi(b)$$

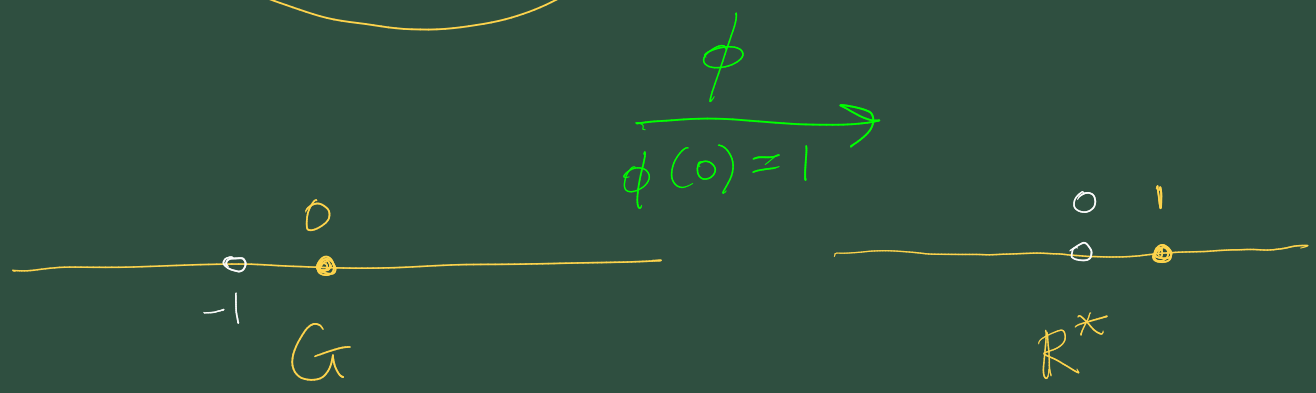
ie. $\phi(a + b + ab) = \phi(a)\phi(b)$.

Consider the result of Q26. namely

$\phi : G \to H$ an iso $\Rightarrow \phi(e_G) = e_H$.

$\phi : G \to \mathbb{R}^*$, $\phi : \mathbb{R}\setminus\{-1\} \to \mathbb{R}\setminus\{0\}$

i.e. $\phi(0) = 1$



Let's try ϕ defined by

$$\forall x \in G \quad \phi(x) = x+1$$

this is certainly a bijection ~~between~~ from $\mathbb{R}\setminus\{-1\} \longrightarrow \mathbb{R}\setminus\{0\}$ and

$\phi(0) = 0+1 = 1$ as required.

We need to prove whether ϕ satisfies the homomorphism property.

$\forall\, a, b \in G = \mathbb{R}\setminus\{-1\}$

$$\phi(a * b) = \phi(a + b + ab)$$
$$= a + b + ab + 1$$

$$= (a+1)(b+1)$$
$$= \phi(a)\phi(b)$$

So ϕ is an isomorphism.

Q2. $\mathbb{C}^* =$ gp of non-zero complex under multiplication.

Claim $\mathbb{C}^* \cong H$

$$H = \left\{ \begin{pmatrix} a & b \\ -b & a \end{pmatrix} : \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \in GL_2(\mathbb{R}) \right\}$$

a subgroup of $GL(2, \mathbb{R})$

note. $\begin{vmatrix} a & b \\ -b & a \end{vmatrix} = a^2 + b^2 \neq 0$.

ie. a, b not both zero.

Find an iso $\phi : \mathbb{C}^* \longrightarrow H$.

Two ways to think about/represent $z \in \mathbb{C}$.

$z = x + iy$, $x, y \in \mathbb{R}$

$= re^{i\theta}$, $r \geq 0$, $0 \leq \theta < 2\pi$

Which to use?

Some guidance for this is to look at demands of homomorphism prop.

$$\forall\, z, w \in \mathbb{C}^* \quad \phi(\underbrace{zw}_{\text{prod in } \mathbb{C}^*}) = \underbrace{\phi(z)\phi(w)}_{\text{prod of matrices}}.$$

Consider

$$\begin{pmatrix} a & b \\ -b & a \end{pmatrix}\begin{pmatrix} c & d \\ -d & c \end{pmatrix} = \begin{pmatrix} ac-bd & bc+ad \\ -cb-ad & ac-bd \end{pmatrix}$$

$$= \begin{pmatrix} ac-bd & cb+ad \\ -(cb+ad) & ac-bd \end{pmatrix}$$

$$(x+iy)(u+iw) = (xu-yw) + i(xw+yu)$$

$$re^{i\theta}\, se^{i\mu} = rse^{i(\theta+\mu)}$$

From these there appear to be similarities between product in Cartesian form and the matrix product

So let's propose a ϕ

$$\phi(a+ib) = \begin{pmatrix} a & b \\ -b & a \end{pmatrix}$$

Check hom prop. It looks right

$$\phi((a+ib)(c+id))$$

$$= \phi(ac-bd+i(ad+bc))$$

$$= \begin{pmatrix} ac-bd & ad+bc \\ -(ad+bc) & ac-bd \end{pmatrix}$$

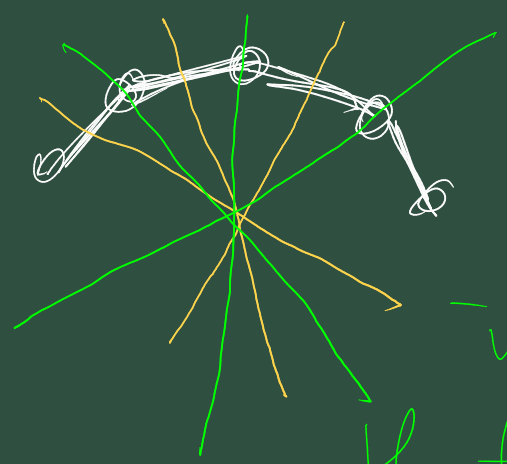
$$= \begin{pmatrix} a & b \\ -b & a \end{pmatrix} \begin{pmatrix} c & d \\ -d & c \end{pmatrix}$$

$$= \boxed{\phi(a+ib)\,\phi(c+id).}$$

Q 12 Prove $S_4 \not\cong D_{12}$.

S_4 = group of all perms of 4 objects

D_{12} = group of all symmetries of a regular dodecagon (12-sided regular polygon).



Idea: To find a property that is preserved by isomorphisms that is differen S_4 and D_{12}. Which will prove there can't be an isomorphism

Consider parts of theorem 9.6

2. $|S_4| = 4!$ $\qquad$ $|D_{12}| = 2 \cdot 12$

$= 24$ $\qquad$ $= 24$.

3. Abelian status?

Both S_4, D_{12} are non-abelian.

4. Cyclic status?

Both S_4, D_{12} are non-cyclic.

5. think about their subgroups.

D_{12}: think about it's subgroup.

$$D_{12} = \{e, r, r^2, r^3, \ldots, r^{11}, s, rs, r^2s, r^3s, \ldots, r^{11}s\}$$

D_{12} has the subgroup

$$R = \{e, r, r^2, \ldots, r^{11}\}.$$

$= \langle r \rangle$, the cyclic subgroup generated by r.

$|R| = |r| = 12$.

If $\phi : D_{12} \to S_4$ is an isomorphism

then S_4 must also have a cyclic subgroup of order/size 12.
i.e. it must have a permutation $\sigma \in S_4$ with $|\sigma| = 12$.

Claim: no such permutation exists in S_4.
Let's detail the orders of elements of S_4.
Think of disjoint cycle representations of elements of S_4.

the identity (1) has order 1
a cycle $(a\ b)$ has order 2.
a 3-cycle $(a\ b\ c)$ has order 3
a 4-cycle $(a\ b\ c\ d)$ " " 4.
a permutation $(a\ b)(c\ d)$ has order 2.

what about $(a\ b\ c)(d)$
$= (a\ b\ c)$

So S_4 has no elements with order > 4, so no elements of

order 12.

Therefore $S_4 \not\cong D_{12}$

<u>Q26</u> A general result.

Let $\phi : G \to H$ be an isomorphism.

Let e_G, e_H be the two identities

Claim $\phi(e_G) = e_H$.

i.e. claim $\forall h \in H \quad h\phi(e_G) = \phi(e_G)h = h$

<u>Proof</u> Let $h \in H$, let's say $\phi(g) = h$, for some $g \in G$

$$\begin{aligned} h\phi(e_G) &= \phi(g)\phi(e_G) \\ &= \phi(ge_G), \text{ by hom-prop of } \phi \\ &= \phi(g), \quad ge = g \\ &= h, \text{ by def of } g. \end{aligned}$$

and similarly can show

$\phi(e_G)h =$

$\vdots$

$= h.$

Therefore $\phi(e_G) = e_H$.

$\boxed{Q34}$ An automorphism is an isomorphism

$$\phi : G \longrightarrow G.$$

ie. it captures the ways a group can represent itself different.

ie. ϕ must be a bijection $G \longrightarrow G$ with the hom prop.

$$\forall\, a, b \in G \quad \phi(ab) = \phi(a)\phi(b)$$

Consider $(\mathbb{C}, +)$ additive group of complex numbers.

$$\text{conj} : \mathbb{C} \longrightarrow \mathbb{C}.$$

$$z \longmapsto \bar{z}$$

in Cartesian terms. $\overline{(a+ib)} = a - ib$

in polar coords $\overline{re^{i\theta}} = re^{i(-\theta)}$

Which coords system is the one to use for addition?

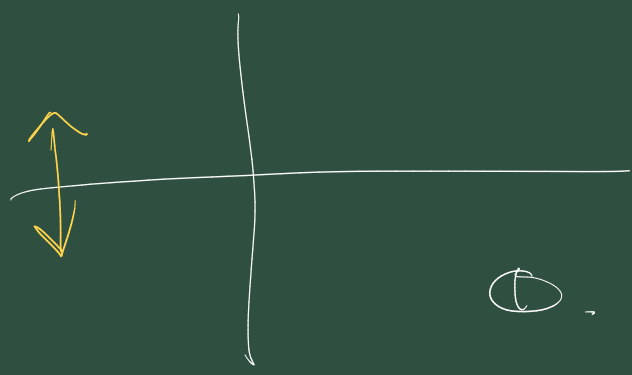
$$(a+ib) + (c+id) = (a+c) + i(b+d)$$

$$re^{i\theta} + se^{r\mu} = ??$$

~~We~~ Use Cartesian form.

$\text{conj}: \mathbb{C} \to \mathbb{C}$.

$a + ib \longmapsto a - ib$.



Clearly this is a bijection.

Does it ~~satisfy~~ ~~satisfy the hom prop?~~

conjugation is reflection in Real axis

$$\cancel{(a+ib) + (c+id) = (a+c) + i(b+d)}$$

$$\cancel{= (a+c) - i(b+d)}$$

$$\cancel{= (a-ib) + (c-id)}$$

$$\cancel{= \overline{(a+ib)} + \overline{(c+id)}}$$

✓

Q35) Same but for $\text{conj}: \mathbb{C}^* \to \mathbb{C}^*$.

Claim: $\overline{zw} = \overline{z}\,\overline{w}$

Proof (using polar form)

$$\overline{re^{i\theta}\, se^{i\mu}} = \overline{rse^{i(\theta+\mu)}}$$

$$= rse^{i(-\theta-\mu)}$$

$$= re^{i(-\theta)}\, se^{i(-\mu)}$$

$$= \overline{re^{i\theta}}\ \overline{se^{i\mu}}.$$